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# Design and Testing of a Nanometer Positioning System

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*This paper presents the design and implementation of a positioning system with a dc servomotor and ball-screw mechanism used to realize high-precision positioning over a wide travel range with nanometer level positioning error and near zero overshoot. Instead of the popular dual-model control strategy and friction compensation, a high-gain proportional-integral-derivative controller is used to realize a single-step point-to-point positioning. The controller parameters are obtained by placing closed-loop poles according to the macrodynamics of a ball-screw mechanism only to avoid identification of microdynamics and friction modeling. In order to suppress the overshoot caused by actuator saturation in long-stroke positioning, a trajectory planning method is applied to calculate the input of the closed-loop system. Experimental and simulation results demonstrate that single-step precision positioning responses to different size commands are achieved without producing any large overshoot. In point-to-point positioning from 100 mm down to 10 nm, the positioning error is within 2 nm and the response dynamics is satisfactory.* [DOI: 10.1115/1.4000811]

*Keywords: nanometer positioning, ball screw, PID controller, saturation*

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## 1 Introduction

The fast development of ultraprecision machining and semiconductor manufacturing technologies in recent years has promoted an increasing demand and research interest in high-precision stages. A positioning mechanism needs not only to achieve a nanometer positioning accuracy but also to reach a few hundred millimeter motion travel.

There is a large variety of actuators used in nanometer positioning instruments. Usually, the piezoelectric actuator is chosen in the design of a nanopositioner, but the motion range is just tens of microns only [1]. Friction drive [2], inchworm mechanism [3], and aerostatic lead screw [4,5] can provide both high resolution and long range with single motion stage, but the speed and acceleration of these mechanisms are confined. These systems also have the disadvantages of complicated structure and high cost. Therefore, a low-cost nanometer positioning stage with a servomotor and ball-screw mechanism is still a hot research objective.

Ball screw is a very popular driving mechanism used in the high-speed and long-stroke precision positioning stage. However, due to the complicated dynamics of friction force arising from the contact interface [6–9], the control of a motion stage with nanometer positioning accuracy has been shown to be very difficult. The so-called dual-model control strategy, two combined controllers designed according to the characteristics of microdynamics and macrodynamics, respectively, is a popular method used to realize precision positioning in the presence of friction [10–13]. However, because the friction parameters of a transmission mechanism are time and position varying, the means of transition from macrodynamics to microdynamics remains vague. The inclusion of friction dynamics as part of the control law does improve the precision positioning, but the controller design becomes time consuming and difficult [13,14]. In addition, the controller design procedure requires sufficient knowledge of both classical and modern control theories. Such knowledge is also required for design of the state feedback controller. The industrial requirements for the design of controllers are not only high positioning performance, but also ease in understanding and designing controller parameters and simple structure. For practical applications, the proportional-integral-derivative (PID) control is a commonly used control structure for its simplicity and reliability [15,16]. One of the possible ways to achieve nanometer accuracy using a PID controller in the ball-screw mechanism is to apply a higher loop-gain [4,17] to suppress the effect of friction.

Therefore, a dc motor and ball-screw-drive positioning system is proposed to satisfy the industrial requirements and to realize high-precision positioning over a wide travel range with nanometer level steady-state error and near zero overshoot, which is one of the main concerns in controller design for such machining processes as microcutting, milling, and turning. For this purpose, a high-gain PID controller is designed according to the macrodynamics of the drive mechanism only and friction is treated as an external disturbance. In order to avoid a large overshoot in long-stroke positioning, the input reference command of the closed-loop system is designed and switched according to the output response state by taking into account the limitations of feed drive. One advantage of doing so is the avoidance of addressing the microdynamics, and another advantage is that the closed-loop system can be designed independently, and its stability and accuracy will not be influenced by the design of the reference input. Moreover, the use of high-gain controller enables the closed-loop system to be insensitive to the variation of friction parameters.

## 2 Feed Drive Stage Specification

As shown in Fig. 1, a dc motor with brushes is used as the main actuator. The forward path of the positioning system begins from the computer equipped with an analog interface, through the amplifier, the dc motor, and the ball screw of C0 grade accuracy, to the table supported on the linear ball guide. In the backward path, an Agilent linear interferometer (10702A) is employed to count the displacement between the interferometer and the retroreflector and to send the digital data to the computer. The basic optical resolution using the linear interferometer is one-half wavelength (0.316 $\mu$m). The system resolution is increased to 1.2 nm ($\lambda/512$) by using an electronic resolution extension with a resolution extension factor of 256 (for Agilent 10897B).

A photograph of the ball-screw drive is shown in Fig. 2. The table is rigidly connected to the nut. The ball screw is connected to the motor by a nonbacklash coupling with high rotational stiff-

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Contributed by the Dynamic Systems Division of ASME for publication in the Journal of Dynamic Systems, Measurement, and Control. Manuscript received May 23, 2008; final manuscript received November 24, 2009; published online February 9, 2010. Assoc. Editor: Won-jong Kim.


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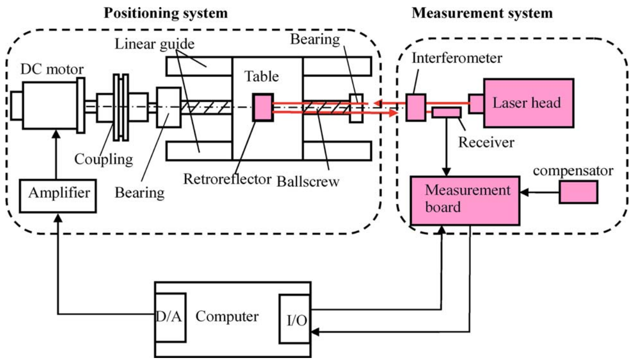


**Fig. 1 Configuration of ball-screw-drive table system**

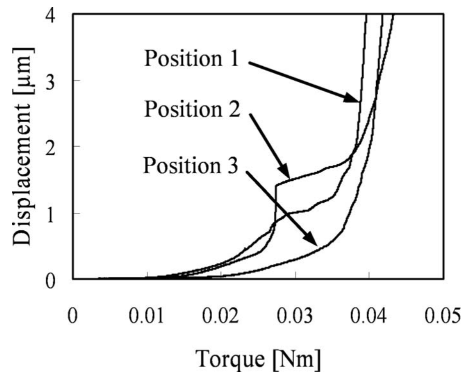


**Fig. 3 Table displacement versus motor torque with $u = 0.05t$ V**

ness and supported by ball bearings on both sides. The effective stroke of the drive mechanism is about 200 mm because of limitation of the guides. The physical parameters of the system are listed in Table 1.

To achieve the highest possible measurement accuracy, the positioning and measurement system is mounted on a vibration-isolated stage for providing sufficient and appropriate isolation of the optical components from the effects of vibration. Laser-interferometer positioning system depends on the laser's wavelength of light (WOL) as its basic scale length. The necessary information for wavelength compensation can be obtained automatically by using the appropriate automatic compensator board and environmental sensors. In the ball screw-drive system, the 10896B laser compensation board is used with the 10897B high resolution laser axis board to compensate for changes in wavelength of light and for changes induced by thermal expansion and to increase the measurement accuracy and repeatability. Air sensor is used to measure temperature and pressure of air and account for humidity for automatic calculation of WOL correction and to send information to the 10896B. Also, the interferometer and retroreflector are placed to minimize Abbé error. The measurement path is directed as close as possible to the actual work area where the measurement process takes place. It is suspected that air turbulence near the laser beam path can cause measurement problems with sporadic counting and drift. The area around the laser beam is shield by one plastic sheet.

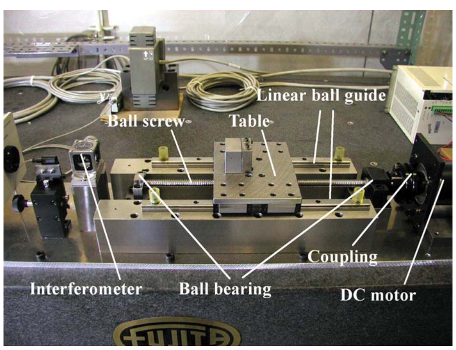


**Fig. 2 Ball-screw-drive positioning system**

**Table 1 Positioning system parameters**

| Description | Parameters | Value |
|---|---|---|
| Inertia of rotating part | $J$ | $3.48\times10^{-4}$ kg m$^2$ |
| Mass of sliding part | $m$ | 2.41 kg |
| Motor resistance | $R_m$ | 4.8 Ω |
| Armature inductance of motor | $L_m$ | 0.6 mhartree |
| Torque constant of motor | $K_i$ | 0.119 V s/rad |
| Back EMF constant of motor | $K_v$ | 0.119 V s/rad |
| Damping coefficient of motor | $C_m$ | $3.88\times10^{-4}$ N m s/rad |
| Damping coefficient of ball screw | $C_n$ | $7.17\times10^{3}$ N s/m |
| Stiffness of ball screw-nut | $K_n$ | $3.2\times10^{8}$ N m |
| Axial stiffness of bearing | $K_b$ | $9.4\times10^{7}$ N m |
| Diameter of ball screw | $D$ | 12 mm |
| Screw length between bearings | $L$ | 355 mm |
| Stiffness of coupling | $K_c$ | $29\times10^{3}$ N m/rad |
| Transmission ratio | $h$ | $4.78\times10^{-4}$ m/rad |
| Friction torque | $T_f$ | 0.04 N m |
| Amplifier gain | $G_m$ | 5.0 |
| Amplifier saturation voltage | $U$ | 3.0 V |

In order to achieve high-precision positioning, it is necessary to preload the ball screw to eliminate the backlash. Ball-screw preload also play an important role in improving the rigidity, accuracy, and life of the positioning stage. However, the preload also produces a significant friction between the ball screw and the nut, which makes it difficult to achieve a nanometer level of resolution. Moreover, there are friction force in the linear guide, bearings, and the motor, and as shown in Fig. 2, the friction force is fairly large.

When a ramp voltage is applied to the motor, an increasing motor current and consequently an increasing torque are generated. Figure 3 shows the table position and motor torque in three repeated tests, each with the table at a different position on the guide.

As shown in Fig. 3, before a friction breakaway, the friction torque is equal to the motor torque and the table has a displacement via deformation. The deformation behaves as a spring with its stiffness $k_f$ being variable with time and position. The test was repeatedly performed 27 times with the table at different positions on the guide. The average value of the breakaway force measured is defined as the friction torque of the drive mechanism $T_f$. However, the quantitative description of the presliding phenomena is highly nondeterministic.

## 3 Drive Mechanism Modeling and Analysis

Figure 4 shows the idealized dynamic model of the drive mechanism, where input voltage $u$ to the amplifier is the control

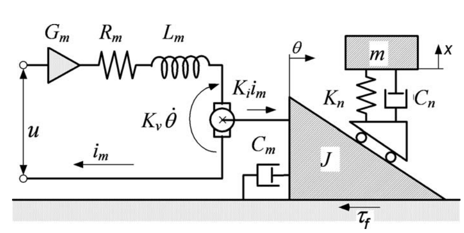


**Fig. 4 Dynamic model of drive mechanism**



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input, and table position $x$ is the only system output, $i_m$ is the motor current, $\theta$ is the angular position of the motor, and the meanings and numerical values of the other symbols are listed in Table 1.

The drive mechanism is simplified as a two-mass system by neglecting the stiffness of bearing and coupling. In comparison with other components, $L$ is so small and $K_n$ is so large that they can be neglected. The plant can thus be modeled as a one-mass system. Let $L_m \rightarrow 0$ and $K_n \rightarrow \infty$; a transfer function from input voltage $u$ to table displacement $x$ can be expressed as

$$H_{\mathrm{macro}}(s) = \frac{X(s)}{U(s)} = \frac{hG_m K_i}{s[(h^2 m + J)R_m s + C_m R_m + K_v K_i]} = \frac{d_0}{s(s + c_1)} = \frac{0.17}{s^2 + 9.517s} \tag{1}$$

The parameters in Eq. (1) are explained in Table 1 along with their numerical values. Equation (1) describes the macrodynamics of the mechanism by neglecting friction and high-order dynamics caused by compliances. In order to achieve high-precision positioning, the servobandwidth of the positioning system must be increased to lower its sensitivity to disturbance. The servobandwidth, however, is mainly limited by the mechanical resonance of the drive mechanism. The finite element analysis done with all the compliances taken into account indicates that the first natural frequency of the drive mechanism is about 623 Hz, which is far beyond the bandwidth of such positioning systems. This testifies that the ball-screw mechanism can be simplified by neglecting the high frequency dynamics as a second-order system with friction nonlinearity.

Because of the friction, the behavior of the system prior to continuous slipping at the friction interface is completely different from that of its macrodynamics. The elastic deformation of the boundary layer dominates the system behavior in the microdynamics. A displacement at the table of several micrometers may occur through deformation until the applied torque exceeds some threshold. If the deformation behaves as a linear spring with its stiffness $k_f$ being constant, and high-order dynamics are neglected, the microdynamics can then be modeled as

$$H_{\mathrm{micro}}(s) = \frac{X(s)}{U(s)} = \frac{d_0}{s^2 + c_1 s + c_0} \tag{2}$$

where $c_0 = k_f/(h^2 m + J)$. This characteristic has a time-varying phenomenon with respect to the operating time, the temperature, and the humidity. It is very difficult to establish an accurate dynamic model by just giving the breakaway force and $k_f$.

## 4 Control Strategy

**4.1 Design of the Closed-Loop Control System.** The purpose of the positioning system is to drive the table from one point to another point as fast as possible with near zero overshoot and nanometer level positioning error. However, the use of ordinary PID control to achieve rapid and precision positioning often fails to result in a large overshoot, so a PID control system with input trajectory planning shown in Fig. 5 is used in experiment.

As shown in Fig. 5, $x_d$ is the step command, $K_c$ is the proportional gain, $T_i$ is the integral time, $T_d$ is the derivative time, and $N$ is the filter factor for the derivative term. The proportional and derivative terms are placed in the backward path. With the same closed-loop poles placement, it gives rise to a much smaller overshoot than a traditional PID controller in step responses. The PID controller used in experiment is originally described in Ref. [4] with parameter calculation and antiwindup techniques for an aerostatic slide system. For a long-stroke positioning system with high-gain controller, the problem of actuator saturation should always be taken into account if a smooth motion without overshoot

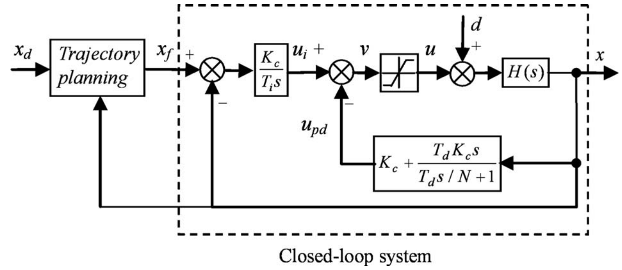


**Fig. 5 Block diagram of positioning system**

has to be achieved. To solve the problem, a trajectory planning element is added between the step command and the closed-loop system. $x_f(t)$ denotes the input trajectory of the closed-loop system.

The closed-loop transfer function from $x_f$ to $x$ can be expressed as

$$G(s) = \frac{X(s)}{X_f(s)} = \frac{K_c d_0[(N+1)T_i T_d s^2 + (NT_i + T_d)s + N]}{T_i T_d s^4 + (T_i T_d c_1 + NT_i)s^3 + [T_i T_d(K_c d_0(N+1) + c_0) + NT_i c_1]s^2 + [NT_i(K_c d_0 + c_0) + K_c d_0 T_d]s + NK_c d_0} \tag{3}$$

The servobandwidth of the positioning system must be increased to lower the sensitivity to disturbance. The use of a high loop-gain is motivated by the unknown parameter $c_0$ in the transfer function of the drive mechanism. If the loop-gain is high enough, i.e., $K_c$ is high enough, the effect of $c_0$ can be neglected and the transfer function can be simplified as follows:

$$G(s) = \frac{X(s)}{X_f(s)} = \frac{K_c d_0[(N+1)T_i T_d s^2 + (NT_i + T_d)s + N]}{T_i T_d s^4 + (T_i T_d c_1 + NT_i)s^3 + [T_i T_d K_c d_0(N+1) + NT_i c_1]s^2 + [NT_i K_c d_0 + K_c d_0 T_d]s + NK_c d_0} \tag{4}$$

It can be seen from above that the closed-loop poles can be arbitrarily placed according to the four controller parameters. Let multiple closed-loop poles be assigned at $-p$, then the controller parameters are found to be

$$\begin{aligned} K_c &= \frac{15p^4 - 4c_1 p^3}{d_0(4p - c_1)^2} \\ T_i &= \frac{15p - 4c_1}{4p^2 - c_1 p} \\ T_d &= \frac{81p^4 - 108c_1 p^3 + 54c_1^2 p^2 - 12c_1^3 p + c_1^4}{(15p^4 - 4c_1 p^3)(4p - c_1)} \\ N &= \frac{81p^4 - 108c_1 p^3 + 54c_1^2 p^2 - 12c_1^3 p + c_1^4}{15p^4 - 4c_1 p^3} \end{aligned} \tag{5}$$

Because of actuator saturation, the problem of integral windup should be considered for a long-stroke positioning system with a pure integrator [18,19]. Otherwise, the closed-loop system may fall into diverging vibration once the operating point is out of a critical range in the state space. A solution for this problem is the antiwindup technique. In this paper, a modified antiwindup integral reset algorithm is used.

At each sampling, control output $v$ is computed according to the basic control algorithm. If the controller output gets saturated, i.e., $|v| > U_s$, then the integrator state is replaced with $u_{pd} + U_s$ (when $|v| > U_s$) or $u_{pd} - U_s$ (when $|v| < -U_s$). The algorithm above is equivalent to an infinite feedback gain and maintains a perfect tracking between $v$ and $u$.



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**4.2 Design of the Deceleration Trajectory.** In a small working range, the step response of the closed-loop system, with the multipoles at $-p$, is free from overshoot because the control signal is always below the actuator saturation. In a large working range, where the saturation characteristic influences the mechanism motion, overshoot becomes unavoidable even with the proposed antiwindup algorithm. In the motion control for computer numerical control machine tools, speed and acceleration control are always used. The reference commands should pass through the predesigned acceleration/deceleration processor before they are fed into the servoloop. Consequently, a smooth motion profile can be obtained. Therefore, it is crucial to identify and integrate the feed drive system limitations with the reference input to achieve the desired responses.

For long-stroke positioning, the controlling process generally includes four stages such as acceleration stage, constant-speed stage, deceleration stage, and positioning stage, and the latter two have an important effect on the positioning accuracy and positioning time [20]. The overshoot caused by actuator saturation cannot be mitigated by simply shaping the closed-loop input by passing one step command through a low-pass filter, because the responses of a low-pass filter to different step heights have the same steady time but not the same constant speed, which should be limited by actuator saturation. The positioning response of a closed-loop system to a larger step height takes longer time than that to a smaller height. In order to suppress the overshoot caused by actuator saturation, an improved trajectory planning method, based on the design of the deceleration stage of positioning, is applied to this system.

Let $x(t)$ be the measured displacement of the table. The input of the closed-loop system is calculated on line according to the following algorithm:

$$X_f(s) = \begin{cases} \dfrac{x_d}{s}, & 0 \le t < t_s \\ \dfrac{x(t_s)}{s} + \dfrac{x_d - x(t_s)}{s} \cdot F(s), & t \ge t_s \end{cases} \tag{6}$$

where $t_s$ denotes the switch point. Simple low-pass filter $F(s)$ is applied to calculate the input of the closed-loop system for the last two positioning stages but not the positioning as a whole. Before the switch point, the input of the closed-loop system is step command $x_d$, after the point it is one specified trajectory generated by the filter $F(s)$ with the switch condition and step height $x_d$ taken into account. Let

$$F(s) = \frac{\omega}{s + \omega} \tag{7}$$

Define the positioning error as $e(t)=x(t)-x_d$. The input of the closed-loop system in the time domain is derived from Eqs. (6) and (7) by reverse Laplace transform:

$$x_f(t) = \begin{cases} x_d, & 0 \le t < t_s \\ x_d - e(t_s)\exp(-\omega(t-t_s)), & t \ge t_s \end{cases} \tag{8}$$

The switch position usually is decided by velocity and positioning error of the table. So a switch function can be defined as follows:

$$f(e) = |e| - \left| \frac{\dot{e}}{\alpha\omega} \right| \tag{9}$$

The sign of quantities $f(e)$ must be continuously monitored. After the sign is changed, the time $t_s$ is determined and the sign of $f(e)$ will not be monitored. At the time $t_s$, the input of the closed-loop system is switched from step command to an exponential mode trajectory calculated by Eq. (8) for decelerating and positioning the table smoothly. $\omega$ decides the theoretical positioning time of the trajectory.

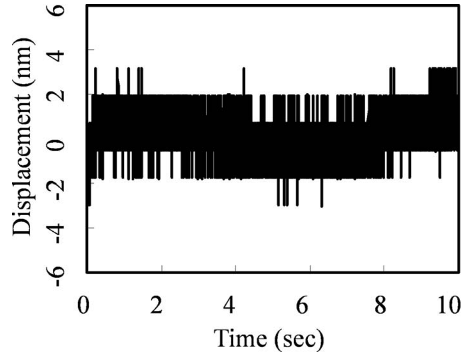


**Fig. 6 Measurement noise**

## 5 Experimental Results and Analysis

**5.1 Determination of Controller Parameters.** The controller is discretized and implemented on a PC. A C programming language running on the DOS platform is used. A 16-bit digital/analog board is used to send out the analog control voltage to the power amplifier. The table position is read, via a parallel input/output board, from the laser interferometer. The sampling period is 0.1 ms. The measurement noise is given when the table is at rest. The laser interferometer output is shown in Fig. 6. The limit value of the noise is about $\pm 3$ nm.

With high loop-gain, the effect of uncertain friction can be neglected, and high loop-gain is always preferred for better closed-loop performance and positioning accuracy. According to some characteristics of the positioning system, such as motor power, amplifier capacity, and the unmodeled high frequency dynamics, the closed-loop multipole $-p=-2\pi\times 50$ is chosen for a closed-loop bandwidth of 50 Hz. The controller parameters calculated by Eq. (6) are listed in Table 2.

In simulation, the integral windup is observed when the step input of the closed-loop system is larger than 0.1 mm and the closed-loop system is not stable without any antiwindup scheme. As shown in Fig. 7, the simulation of 10 mm step response of the closed-loop system is stable in the simulation with the antiwindup scheme. Very shortly after the input is applied, the integrator integrates its inputs and reaches a very large value. After this, the proposed antiwindup algorithm sets $u$ at the hardware limit. The system moves to the setpoint at the largest acceleration. Before the system reaches the setpoint, the moving velocity is very high and has to be reduced. The required deceleration is too large to be achieved even when $u$ reaches the negative hardware limit. An overshoot appears. The overshoot is relatively larger and $u$ does reach the hardware limit again. Then the negative overshoot appears and it is very small, and $u$ does not reach the hardware limit thereafter. Finally the system stays at the setpoint.

With the trajectory planning method described above, the control algorithm divides the positioning responses into two phases according to the inputs of the closed-loop system. For a long-stroke positioning task, the exponential trajectory in Eq. (8) for the deceleration described by $x_f(t)$ is characterized by parameters $\omega$ and $\alpha$. The largest value of $\omega$ is limited by the amplifier saturation. Select the values of $\omega$ and $\alpha$ according to the simulation results so that a smooth transition can be realized between the two phases of positioning:

**Table 2 Parameters of PID controller**

| Poles | $K_c$ | $T_i$ | $T_d$ | $N$ |
|---|---|---|---|---|
| $-2\pi\times 50$ | $5.49\times 10^5$ | $1.19\times 10^{-2}$ | $4.19\times 10^{-3}$ | 5.23 |

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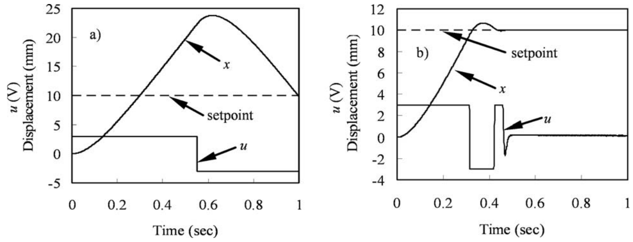


**Fig. 7 Simulation responses of closed-loop system to 10 mm step input: (*a*) without antiwindup scheme and (*b*) with antiwindup scheme**

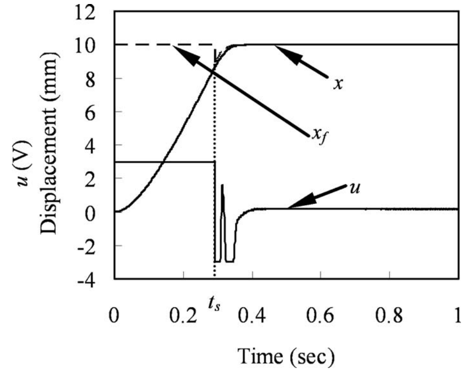


**Fig. 8 Simulation of 10 mm positioning response**

$$\alpha = 0.5, \quad \omega = 60$$

As shown in Fig. 8, the system moves to the reference point with the largest acceleration at the beginning. Before the system reaches the setpoint, a switch of the closed-loop input is automatically executed according to the motion state described by Eq. (9). Then, the system brakes with the largest deceleration because the control error, i.e., the difference between the input trajectory and the table displacement, is suddenly reduced. If the positioning distance is long enough, the system should reach its peak speed limited by actuator saturation and system damping. Defined by Eq. (9), after switch point, the control error curves are very similar due to the same switch velocity, i.e., the peak speed. The theoretical peak speed is 53.6 mm/s calculated using the parameters in Table 1. Because the values of $\omega$ and $\alpha$ are selected with the actuator saturation taken into account, $u$ does not reach the hardware up-limit thereafter and the input trajectory leads the table to the reference position without an overshoot.

As shown in Fig. 8, the plant input function defined by Eq. (8) is piecewise continuous because it is continuous at all but a point at time $t_s$ and the left and right hand side limits exist at the point. Because the plant input meets the assumption of the closed-loop control design, the closed-loop system with minus multiple poles remains stable even the plant input undergoes such abrupt changes.

Without the deceleration trajectory element, the step responses

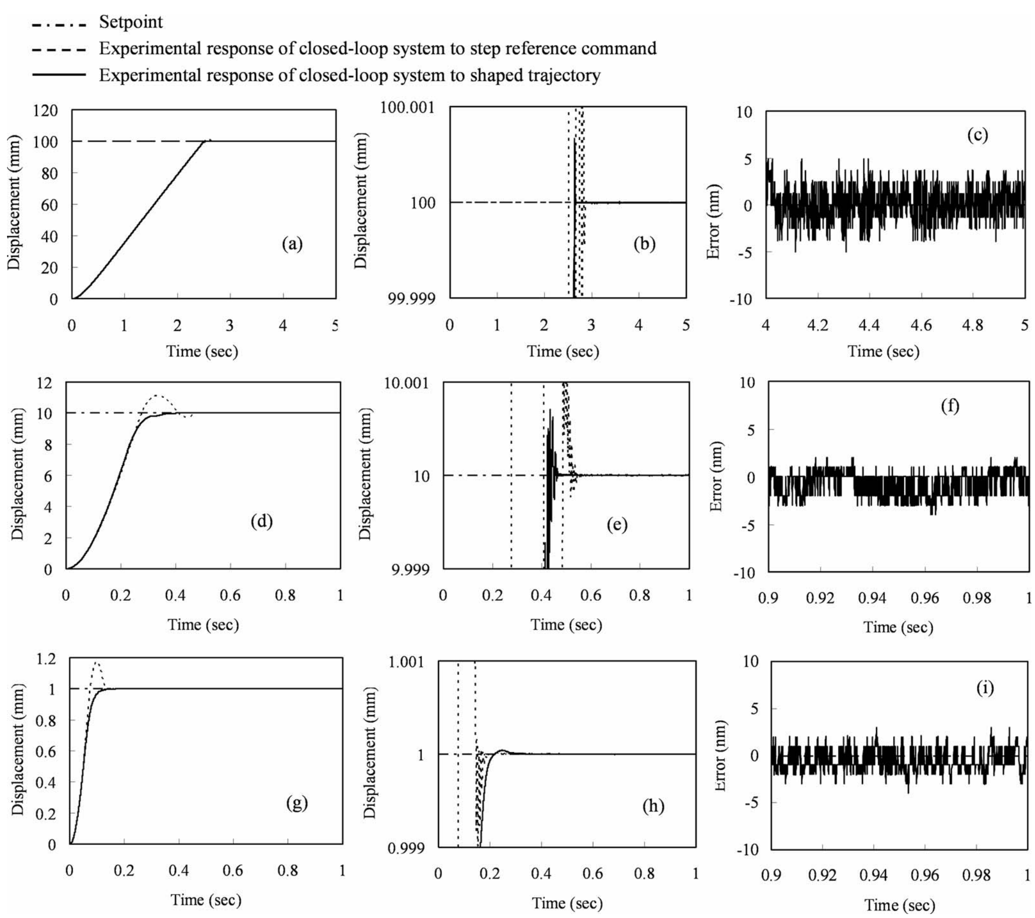


**Fig. 9 Experimental responses: (*a*) 100 mm positioning response, (*b*) zoomed view of (*a*), (*c*) steady-state error of 100 mm positioning response, (*d*) 10 mm positioning response, (*e*) zoomed view of (*d*), (*f*) steady-state error of 10 mm positioning response, (*g*) 1.0 mm positioning response, (*h*) zoomed view of (*g*), and (*i*) steady-state error of 1.0 mm positioning response**

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of the closed-loop system can be achieved without an undesirable overshoot if the step height is smaller than 0.1 mm. In experiments, the deceleration trajectory is only added when step command $x_d$ is larger than 0.1 mm, otherwise $x_f$ is equal to $x_d$.

**5.2 Experimental Results.** Experimental responses corresponding to 100 mm, 10 mm, and 1 mm step command positioning are shown in Fig. 9. The dashed line represents the responses of the closed-loop system to step reference commands and due to actuator saturation large overshoot that appeared. With the deceleration trajectory planning method, the overshoot due to amplifier saturation is avoided and smooth positioning responses are achieved. To clearly visualize the data, the results of transient responses near the targets and steady-state error responses are enlarged and shown separately due to the large difference in data magnitudes. The steady-state error responses demonstrate good precision performance at the nanometer level. The peaks in magnitude may come from the noise of measurement devices and cannot be distinguished in our case. Due to the same switch velocity and control error curve, overshoots, which appeared in 10 mm and 100 mm responses, are at the same level.

To explore the repeatability of the positioning control, the stepwise reference commands from 10 nm to 0.1 mm also were performed in the experiments. The average error from 0.9 s to 1.0 s (from 4.9 s to 5.0 s for 100 mm) is defined as the steady-state error for the purpose of judging position accuracy. The steady-state error is less than 2 nm regardless of the motion range. In the results above, each step command was repeatedly performed with the table at different positions on the guide. Since the table positions are different and due to the varying nature of the friction, there will be uncertainties over the whole stroke. With a sufficiently high-gain controller, the transient responses are much less affected by those uncertainties and precision positioning can thus be achieved.

## 6 Conclusions

A long-range precision positioning system with a dc motor and ball-screw mechanism is designed to achieve the nanometer level steady-state position accuracy and near zero overshoot. The system macrodynamics is developed and simplified. A high-gain PID controller is designed to control the stage based on the system macrodynamics by neglecting friction and high-order dynamics. In order to suppress overshoot that appeared in millimeter level step responses due to actuator saturation, the closed-loop input is automatically switched from step command to a deceleration trajectory by taking into account the actuator saturation and motion state. The closed-loop bandwidth is increased to 50 Hz. This gives a sufficiently high loop-gain to suppress the nonlinear stick-slip behavior of the ball screw and servomotor system and achieves nanometer positioning accuracy. The experimental results show that the single ball-screw positioning system can reach more than 100 mm stroke and the steady-state positioning error is always less than 2 nm and without any large overshoot. The nanometer positioning stage has the advantage of low cost, long range, and simple mechanism installation. The obvious advantage of the control strategy over prevailing technologies is the avoidance of addressing the microdynamics. The positioning stage is very easy for industrial application.

## Acknowledgment

This research is sponsored by the Development Program for Outstanding Young Teachers in Harbin Institute of Technology (HITQNJS.2008.013) and Postdoctoral Foundation of Heilongjiang Province (LBH-Z0713).

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